

F1 PHYSICS-BASED PREDICTIVE PIPELINE

Complete Technical Reference

Layer 1	Layer 2	Layer 3	Layer 4	Layer 5
Aero	Alignment	Racing Line	Tyre Model	Strategy

Telemetry → Aero → Aligned Data → Racing Line → Tyre Degradation → Race Strategy

Data Sources: FastF1 · OpenF1 API · Ergast API · OpenStreetMap

Primary Library: FastF1 (Python) | 2018–2024 seasons

Output: Probabilistic pit strategy · Compound choice · Undercut/Overcut viability · Risk-ranked recommendations

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1. Project Overview & Core Pipeline

This project transforms publicly available Formula 1 telemetry and timing data into a physics-based predictive system. The pipeline is strictly layered — each layer’s output feeds the next as a distribution, not a point estimate, so uncertainty accumulates honestly rather than being hidden. The final output is a probabilistic race strategy recommendation with risk ranking.

Pipeline Dependency Graph

Dependency	Data Passed
L1 → L3	Cd, Cl posteriors feed speed profile aero-corrected normal force
L2 → L3	Aligned distance-grid telemetry feeds curvature and speed profile
L3 → L4	Racing line curvature and speed profile feed lateral/braking energy
L3 ↔ L4	Circular dependency — resolved by iterative refinement (3–5 passes)
L4 → L5	Degradation posteriors, cliff survival probabilities, warm-up curves
L1→5	All posterior distributions jointly sampled per Monte Carlo scenario

Final Outputs

- Optimal compound sequence and pit lap recommendations
- Expected race time distribution across 10,000 simulated scenarios
- CVaR-adjusted risk ranking of all viable strategies
- Undercut viability: P(undercut viable) and minimum gap threshold g^*
- Overcut viability updated every 5 laps
- Tyre cliff warning: P(cliff within k laps)
- SC-robust strategy flags
- No-SC vs SC scenario branch comparison

2. Data Sources & Global Limitations

Data Sources

Source	Coverage	Used For
FastF1	2018–present	Telemetry, lap timing, weather, stint data, track status
OpenF1 API	2023–present	Enhanced real-time session data, stint granularity
Ergast API	2003–present	Historical lap times, pit stops, results, compounds
OpenStreetMap	Current	Circuit geometry, track boundaries, corner radii
F1 Official Site	Current	Official circuit lengths, DRS zone locations

What FastF1 Does NOT Provide

- Tyre surface or carcass temperature (per tyre)
- Tyre pressure (per tyre)
- Ride height (front or rear)
- Real fuel load per lap (modelled as linear depletion)
- Brake pressure magnitude (binary on/off only)
- Per-tyre load distribution
- Camber and toe angles
- Aerodynamic balance (front/rear downforce split)
- Engine power curve (approximated from RPM + gear)

Telemetry Channel Reference

Channel	Native Rate	Handling	Used In
Speed	~240 Hz	Savitzky-Golay filtered	All layers
Throttle %	~240 Hz	Raw	L2, L4
Brake (bool)	~240 Hz	Nearest-neighbour only	L2, L4
DRS (bool)	~240 Hz	Nearest-neighbour only	L1, L2
RPM / Gear	~240 Hz	Linear interp	L1
X, Y, Z (GPS)	~4 Hz	Dead reckoning upsample	L2, L3
Weather	~1/min	Hold-last + GP uncertainty	L1, L4

3. Uncertainty Propagation Framework

Every layer produces distributions, not point estimates. These distributions are sampled jointly at the start of each Monte Carlo scenario in Layer 5, preserving correlation structure across the full pipeline. Sensitivity analysis identifies which upstream uncertainties dominate the final strategy recommendation.

Uncertainty Budget by Layer

Layer	Uncertainty	Source
L1 — Aero	Cd ± 15–20%, Cl ± 20–25%	Gaussian posteriors from energy method
L2 — Alignment	Segment quality flags	Consistency check pass/fail per segment
L3 — Racing Line	Speed profile GP residual	Fitted to qualifying telemetry residuals
L4 — Tyre	Degradation curve posterior	MCMC posterior from hierarchical model
L4 — Tyre	Cliff timing Weibull	Survival model P(cliff within k laps)
L5 — Strategy	Pit stop time distribution	Per-team empirical + heavy tail
L5 — Strategy	SC occurrence Poisson	Non-homogeneous, circuit specific

Propagation Method

At each of 10,000 Monte Carlo scenarios, one sample is drawn from every upstream posterior and held fixed for the entire race simulation. This means each scenario represents a self-consistent draw from the full joint uncertainty of the pipeline. Strategy rankings are stable when checked between 5,000 and 10,000 scenarios.

$$Score_s = w1 * T_s_mean + w2 * CVaR_{0.8}(T_s) \text{ [} w1=0.7, w2=0.3 \text{]}$$

4. Layer 1 — Aero Parameter Estimation

Estimates circuit-specific aerodynamic drag (Cd) and lift (Cl) coefficients as posterior distributions using a sequential, track-aware pipeline. Energy methods replace instantaneous force methods to avoid noise amplification from acceleration signals. DRS acts as a natural experiment providing an independent Cd calibration check.

Step 1 — Preprocessing

- Compute air density: $\rho = P / (R_d \cdot T)$, $R_d = 287.05 \text{ J/kg}\cdot\text{K}$
- Apply Savitzky-Golay filter to speed channel (window tuned per circuit)
- Compute road gradient theta from Z channel differentiated over distance
- Flag DRS transitions — exclude 0.3s buffer around open/close events
- Exclude SC, VSC, red flag, pit-in/out laps via FastF1 track status channel
- Correct distance channel drift using official F1 circuit length as anchor

Step 2 — Cd Estimation (Energy Method)

On DRS-closed full-throttle straights, energy balance over entry/exit speeds:

$$(1/2) \cdot m \cdot (v_2^2 - v_1^2) = W_{\text{engine}} - C_d \cdot (1/2) \cdot \rho \cdot A \cdot \int (v^2 \, ds) - F_{rr} \cdot ds - m \cdot g \cdot dh$$

- m = car mass ~798 kg (fuel-corrected: 1.8 kg/lap burn)
- A = reference area fixed at 1.5 m^2
- W_{engine} approximated from RPM + gear (bounded $\pm 8\%$ uncertainty)
- F_{rr} = rolling resistance prior ~500–600 N ($\pm 15\%$)
- dh = elevation change from Z channel

Aggregate Cd estimates across multiple straights and laps → fit Gaussian: $C_d \sim N(\mu_{Cd}, \sigma_{Cd})$

Step 3 — DRS Calibration Check

Compare DRS-open vs DRS-closed acceleration profiles on the same straight across laps. Expected delta: $\Delta C_d = 0.08\text{--}0.12$ from published literature. If energy-method estimate falls outside this range, σ_{Cd} is widened accordingly.

$$\Delta C_d = C_{d_closed} - C_{d_open} \approx 0.08\text{--}0.12$$

Step 4 — Cl Estimation (Sequential, Cd Fixed)

In high-speed corners with constant speed (speed variance < 2 km/h):

$$C_l = 2 \cdot (m \cdot v^2 / r - F_{\text{mechanical}}) / (\rho \cdot A \cdot v^2)$$

Mechanical grip component $F_{\text{mechanical}}$ calibrated from slow corners ($v < 80 \text{ km/h}$) where aero contribution is negligible. Aggregate across corners → $C_l \sim N(\mu_{Cl}, \sigma_{Cl})$.

Step 5 — Track-Type Routing

Circuit Class	Cd Source	Cl Source	Notes
Low drag (Monza, Montreal)	Energy method — excellent signal	Limited high-speed corners	Best Cd calibration

High speed (Spa, Silverstone)	Energy method — good	High-speed corners — good	Best for both
Street (Monaco, Baku)	Poor — use Spa/Silverstone prior	Slow corners only — poor	Hold params, widen sigma x1.5
Mixed (Bahrain, Abu Dhabi)	Moderate signal	Moderate signal	Estimate fresh, wider CI

Layer 1 Output

$Cd \sim N(\mu_Cd, \sigma_Cd)$ $CI \sim N(\mu_CI, \sigma_CI)$

Distributions (not point estimates) passed forward to Layers 3 and 5.

Documented Assumptions

- Car mass known within ±2 kg (linear fuel model, 1.8 kg/lap)
- Reference area fixed at 1.5 m^2 (published F1 approximation)
- Engine power approximated from RPM/gear (±8% bounded uncertainty)
- Rolling resistance treated as prior (±15%)
- DRS delta_Cd literature range used as calibration bound, not hard constraint

5. Layer 2 — Telemetry Alignment

Aligns multi-driver, multi-session telemetry onto a clean consistent distance-based spatial grid using physics-constrained DTW on event-segmented signals, with drift-corrected distance reference and channel-aware interpolation.

Step 1 — Distance Channel Drift Correction

$$\text{drift} = d_{\text{raw}}(t_{\text{end}}) - L_{\text{official}}$$

$$d_{\text{corrected}}(t) = d_{\text{raw}}(t) - (t / T_{\text{lap}}) * \text{drift}$$

Official circuit length L_{official} used as anchor. Laps deviating >15m from L_{official} are flagged as corrupted (SC, pit, track limits) and excluded.

Step 2 — Lap Boundary Cleaning

- In-laps and out-laps (pit entry/exit)
- SC, VSC, red flag laps (FastF1 track status channel)
- Race lap 1 (standing start anomaly)
- Qualifying out-laps and cool-down laps

Official sector timestamps used as anchor points on the distance grid.

Step 3 — GPS Upsampling via Dead Reckoning

Upsample 4Hz GPS to ~240Hz between consecutive fixes:

$$x(t+n*dt) = x(t) + \sum_i [v(t+i*dt) * \cos(\psi_i) * dt]$$

$$y(t+n*dt) = y(t) + \sum_i [v(t+i*dt) * \sin(\psi_i) * dt]$$

Position resets to true GPS fix at each new observation. Maximum positional error within a single 250ms GPS interval at 300 km/h: ~0.35m.

Step 4 — Event Segmentation

Segment Type	Detection Rule	DTW Behaviour
Braking zone	Brake=TRUE, speed decreasing	Align on deceleration profile
Corner apex	Speed local minimum ± 5 km/h	Align on speed minimum position
Full throttle	Throttle >95%, brake=FALSE	Align on acceleration profile
Transition	Everything else	Linear interpolation, no DTW

Step 5 — Physics-Constrained DTW

Constraint 1 — Cross-validated window

$$w^* = \text{argmin}_w (1/N) * \sum_i |t_{\text{sector_aligned}} - t_{\text{sector_official}}|$$

Window size minimising sector time recovery error across 10+ laps per circuit.

Constraint 2 — Physics upper bound guard

$$w_max = 2.0 * v_segment_mean \text{ [2s maximum inter-driver offset]}$$

$$w = \min(w^*, w_max)$$

If $w^* > w_max$, segment flagged as genuine driving difference — not force-aligned.

Step 6 — Multi-Driver Consistency Check

$$\epsilon_{ABC} = || \text{align}(A \rightarrow C) - \text{align}(A \rightarrow B) \circ \text{align}(B \rightarrow C) ||$$

Threshold ~0.5m. Failures logged — not silently passed to Layer 3.

Step 7 — Channel-Specific Interpolation (1m distance grid)

Channel	Method	Reason
Speed, X, Y, Z	PCHIP	Smooth first derivative, no oscillations
Throttle %	PCHIP	Continuous signal
RPM	Linear	High native rate, linear sufficient
Brake (bool)	Nearest neighbour	Never interpolate a boolean
DRS (bool)	Nearest neighbour	Never interpolate a boolean
Gear	Nearest neighbour	Discrete signal

Step 8 — Weather Channel

Primary: hold-last-known-value per lap (physically defensible — weather does not un-happen). Uncertainty: Gaussian Process with squared exponential kernel fitted to session observations.

$$k(t, t') = \sigma^2 * \exp(-(t-t')^2 / (2 * l^2))$$

Length scales: $l = 5\text{min}$ (track temp), $l = 15\text{min}$ (air temp). GP posterior uncertainty propagated to Layer 1 air density and Layer 4 tyre baseline.

Documented Assumptions

- Official circuit length accurate to $\pm 1\text{m}$
- Maximum inter-driver timing offset bounded at 2.0s for window guard
- Dead reckoning positional error bounded at ~0.35m per GPS interval at max speed
- Weather GP length scales: 5min track temp, 15min air temp
- 1m distance grid sufficient for Layer 3 curvature resolution

6. Layer 3 — Optimum Racing Line Engine

Reconstructs track geometry using B-spline/clothoid fitting (not raw GPS differentiation), optimises the minimum-curvature racing line within OSM-sourced track boundaries, and generates a theoretical speed profile via CVXPY convex optimisation. Iterative refinement with Layer 4 resolves the circular traction-circle dependency.

Step 1 — Track Geometry via B-spline / Clothoid Fitting

Curvature is NOT computed by double-differentiating GPS points (numerically unstable). Instead, fit a parametric B-spline to the aggregated median position across all drivers and sessions. Curvature is computed analytically from the spline control points.

$$\kappa = (x'y'' - y'x'') / (x'^2 + y'^2)^{3/2} \text{ [from spline, not raw GPS]}$$

Clothoid segments (Euler spirals) used at corner entries and exits — physically motivated as real circuit geometry is designed to clothoid standards. OSM circuit geometry registered to telemetry coordinate frame for validation and boundary extraction.

Step 2 — Track Width & Boundary Extraction

Primary: OpenStreetMap track edges registered to telemetry coordinate frame.

Validation: lateral spread of all drivers across full race weekend gives empirical lower bound on track width (typically within 0.5–1m of true edge).

Kerb detection: Z-channel anomalies identify kerb events → used as boundary anchors at high-usage corners.

Step 3 — Racing Line Optimisation

Solve for the minimum curvature racing line within track boundaries. This is the geometric optimum — computationally tractable and a well-defined problem. It is explicitly documented as a geometric approximation to the true minimum-time optimal control problem.

$$\min \int \kappa(s)^2 ds \text{ subject to: line within track boundaries}$$

Late-session laps only (Q3 final runs, final race stints) used for geometry — track is fully evolved and driver lines are closest to physical limit.

Step 4 — Point-Mass Vehicle Model

Traction limit at any point on the racing line:

$$F_{\max} = \mu * (m * g + (1/2) * \rho * A * C_l * v^2)$$

The aero downforce term (C_l from Layer 1) makes the traction circle speed-dependent — grip increases with speed, capturing the most important vehicle dynamics effect. μ calibrated empirically from qualifying telemetry (Step 5).

Step 5 — Empirical Calibration Against Qualifying

Find μ such that predicted speed profile matches actual fastest qualifier's speed profile within tolerance. This absorbs unknown vehicle parameters (brake bias, suspension geometry, weight transfer) into a single effective friction coefficient per circuit.

$$\mu^* = \operatorname{argmin}_{\mu} \| v_{\text{predicted}}(s; \mu) - v_{\text{actual_Q}}(s) \|^2$$

Step 6 — Speed Profile via CVXPY Convex Optimisation

Forward-backward speed profile using convex optimisation:

Backward pass: max arrival speed at each apex given braking limits

Forward pass: max speed at each exit given traction limits

$v_profile(s) = \min(v_forward(s), v_backward(s))$

Speed-dependent aero traction limit from Layer 1 CI posterior feeds directly into the constraint set. CVXPY handles the constrained optimisation cleanly.

Step 7 — Iterative Refinement with Layer 4

- [0] Iteration 0: Fixed nominal μ (point-mass, constant, no temperature effect)
- [1] Generate racing line and speed profile
- [2] Feed load history (lateral force, braking energy) to Layer 4
- [3] Layer 4 returns updated grip coefficient $\mu(E)$ as function of energy dissipation
- [4] Recompute speed profile with updated traction limits
- [5] Repeat until speed profile convergence (typically 3–5 iterations)

Validation

Compare predicted speed-distance profile against actual fastest qualifier. Target: predicted apex speed within ± 5 km/h of actual. Gap between predicted optimal and actual driver speed quantifies driver performance margin — a useful output in itself, not a validation failure.

Documented Assumptions

- *Geometric racing line is a documented approximation to true minimum-time optimal control*
- *Point-mass model — weight transfer and suspension dynamics are second-order corrections*
- *μ calibrated per circuit from qualifying; may not reflect wet or degraded conditions*
- *OSM boundary accuracy assumed within $\pm 1m$ of true track edge*
- *Late-session laps assumed to represent fully-evolved track conditions*

7. Layer 4 — Tyre Thermodynamics & Wear

Models tyre wear state, grip decay, and cliff probability using cumulative energy dissipation as the wear state variable — replacing an underdetermined full thermal simulator with a physically motivated proxy. A hierarchical Bayesian degradation model fitted across multiple circuits and seasons provides compound-specific degradation curves with honest uncertainty.

Step 1 — Circuit Dominance Classification

$$D = [\text{sum_L}(\kappa_{\text{L}} * v^2) - \text{sum_R}(\kappa_{\text{R}} * v^2)] / [\text{sum}(\kappa * v^2)]$$

$D > 0$: left-dominant (e.g. Silverstone). $D < 0$: right-dominant (e.g. Bahrain). Identifies the most-stressed front tyre. Stress multiplier $\alpha_{\text{dominant}} = 1.15\text{--}1.25$ scales energy dissipation for that tyre.

Step 2 — Energy Dissipation Estimation

$$E_{\text{brake}} = \text{integral}[F_{\text{brake}}(s) * v(s) \, ds]$$

$$F_{\text{brake}} = m * |a_{\text{decel}}| - F_{\text{drag}} - F_{\text{rr}}$$

$$E_{\text{lateral}} = \text{integral}[F_{\text{lateral}}(s) * v_{\text{slip}}(s) \, ds] \quad [F_{\text{lat}} = m * v^2 * \kappa_{\text{appa}}]$$

$$E_{\text{traction}} = \text{integral}[F_{\text{traction}}(s) * v_{\text{slip_long}}(s) \, ds]$$

$$E_{\text{total}}(n) = \alpha_{\text{dominant}} * \text{sum}_{i=1}^n (E_{\text{brake}} + E_{\text{lateral}} + E_{\text{traction}})_i$$

Uncertainty $\sigma_E(n)$ propagated via standard error propagation from noisy speed gradient and Layer 1 aero uncertainty.

Step 3 — Lap Time Decomposition

$$\text{delta_t_lap}(n) = f_{\text{fuel}}(n) + f_{\text{track}}(n) + f_{\text{tyre}}(E_{\text{total}}(n)) + \text{epsilon}(n)$$

Components:

$$f_{\text{fuel}}(n) = -k_{\text{fuel}} * m_{\text{dot_fuel}} * n \quad [k_{\text{fuel}} = 0.03 \text{ s/kg}, m_{\text{dot}} = 1.8 \text{ kg/lap}]$$

$$f_{\text{track}}(n) \sim \text{GP}(0, k_{\text{track}}) \quad [\text{exponential saturation kernel, shared across drivers}]$$

$$f_{\text{tyre}}(E) = \text{delta_t_lap} - f_{\text{fuel}} - f_{\text{track}} \quad [\text{isolated tyre signal}]$$

Late-session laps (Q3, final race stints) used where $f_{\text{track}} \approx 0$, simplifying decomposition.

Step 4 — Hierarchical Bayesian Degradation Model

$$f_{\text{tyre}}(E|\theta) = \beta_0 + \beta_1 * E + \beta_2 * E^2 + \gamma * 1[E > E_{\text{cliff}}]$$

Hierarchical structure across circuits and compounds:

$$\beta_1^{(c,k)} \sim N(\mu_{\beta_1}^{(k)}, \sigma_{\beta_1}^{(k)})$$

Compound-level mean shared across all circuits. Circuit-level deviates from mean.

Physically motivated priors:

- $\beta_1 > 0$ (degradation monotone in energy — tyres don't improve with use)
- E_{cliff} within plausible stint window
- $\gamma > 0$ (cliff always increases lap time)
- Compound ordering: $\beta_1_{\text{C5}} > \beta_1_{\text{C4}} > \dots > \beta_1_{\text{C1}}$ (softer = faster degradation)

Calibration: Qualifying multi-run lap time delta (same tyre set, run 1 vs run 2) used as primary compound calibration — least confounded signal in public data.

Fitting: PyMC MCMC across all available races (2018–2024) for each circuit-compound combination.

Step 5 — Tyre Cliff Survival Model

$$S(E) = P(\text{no cliff before } E) = \exp(-\int_0^E h(e) \, de)$$

$$h(E) = (\beta/\lambda) * (E/\lambda)^{(\beta-1)} \text{ [Weibull hazard]}$$

$$P(\text{cliff within } k \text{ laps}) = 1 - S(E + k * E_{\text{lap_mean}}) / S(E)$$

Cliff events identified from historical data as lap time jumps >1.5s not explained by SC, traffic, or weather.

Step 6 — Post-Pitstop Warm-up Model

$$\mu(n_{\text{out}}) = \mu_{\text{cold}} + (\mu_{\text{opt}} - \mu_{\text{cold}}) * (1 - \exp(-n_{\text{out}} / \tau))$$

$$\tau = \tau_0 * C_{\text{hardness}} / (T_{\text{track}} - T_{\text{ambient}})$$

$\mu_{\text{cold}} \approx 0.75 * \mu_{\text{opt}}$ (literature approximation). C_{hardness} index: C1=5, C2=4, C3=3, C4=2, C5=1. Warm-up curve feeds Layer 5 out-lap time prediction for undercut/overcut calculations.

Step 7 — Cooling Model (Speed-Dependent)

$$E_{\text{effective}}(n) = E_{\text{total}}(n) * (1 - \eta_{\text{cool}})$$

Circuit Class	Avg Speed (km/h)	Cooling Rate	Effect
Monza / Montreal	~230	High	Low η_{cool}
Spa / Silverstone	~210	Medium-high	Medium-low η_{cool}
Bahrain / Abu Dhabi	~195	Medium	Medium η_{cool}
Monaco / Singapore	~160	Low	High η_{cool}

Layer 4 Output Summary

Output	Form	Fed To
Grip coefficient $\mu(n)$	Posterior distribution	L3 refinement, L5
Cumulative energy $E_{\text{total}}(n)$	Point estimate + σ_E	L5
Cliff probability $P_{\text{cliff}}(n,k)$	Probability distribution	L5 Monte Carlo
Out-lap grip $\mu(n_{\text{out}})$	Warm-up curve	L5 undercut/overcut
Degradation curve $f_{\text{tyre}}(E)$	Posterior predictive	L5 strategy window

Documented Assumptions

- Most-stressed-tyre approximation with $\alpha_{\text{dominant}} = 1.15\text{--}1.25$
- Fuel burn linear at 1.8 kg/lap, lap time sensitivity 0.03 s/kg
- Lateral slip angle approximated from path-heading deviation
- Cooling modelled as energy retention modifier, not explicit temperature tracking
- Compound ordering constraint encoded as prior (softer = faster degradation)

- *Cliff identification threshold: >1.5s single-lap jump not explained by external events*
- *Warm-up cold grip = 0.75 x optimal grip (literature approximation)*
- *Track evolution modelled as GP with exponential saturation kernel*

8. Layer 5 — Monte Carlo Strategy Engine

Simulates 10,000 race scenarios by jointly sampling all upstream posterior distributions, models safety car as a non-homogeneous Poisson process, uses historical opponent strategy archetypes and 2-player Nash equilibrium for undercut decisions, and solves for optimal pit strategy via receding horizon dynamic programming updated every 5 laps.

Step 1 — Race State Initialisation (Laps 1–3)

Laps 1–3 are NOT physics-modelled. Empirical approach:

$P(\Delta p_i | p_{grid_i})$ from historical 2018–2024 race data

$P(SC_{lap1}) = 3 * \lambda_{base_circuit}$ [elevated first-lap SC rate]

Race state s_4 = (position, gap, E_{tyre} , stint_n, compound, SC_{status}) initialised at lap 4.

Step 2 — Joint Upstream Uncertainty Sampling

Per scenario j , draw one consistent sample from all upstream posteriors:

$Cd^j \sim N(\mu_{Cd}, \sigma_{Cd}), Cl^j \sim N(\mu_{Cl}, \sigma_{Cl})$

$v_{profile}^j(s) = \bar{v}(s) + \epsilon_v(s), \epsilon_v \sim GP(0, k_v)$

$\beta^j, E_{cliff}^j \sim P(\beta, E_{cliff} | data)$ [Layer 4 MCMC posterior]

$t_{pit}^j \sim Empirical_team \text{ OR } Empirical_team + U(3,8)$ [w/ prob p_{issue}]

All samples held fixed for entire race simulation — preserves inter-layer correlation.

Step 3 — Compound Strategy Space

Dominance pruning:

Strategy A dominates B if total race time $A \leq B$ across all plausible degradation scenarios. Typically reduces search space by 40–60%.

Vectorised evaluation:

S in $R^{(N_{strategies} \times N_{laps})}$ evaluated via NumPy broadcasting

Degradation model pre-computed as lookup table over (compound, energy dissipation) pairs.

Step 4 — Safety Car Model

$\lambda(n) = \lambda_{base}(c) * \phi(n)$

Lap Range	$\phi(n)$ Multiplier	Rationale
Laps 1–3	3.0x	Historically elevated crash rate
Laps 4 – 80% race	1.0x	Baseline mid-race rate
Final 20% laps	1.3x	Increased risk on worn tyres

Scenario branching: run parallel no-SC and SC-at-lap- $N/3$ branches. Strategies robust to both branches flagged as highest-confidence recommendations.

Step 5 — Opponent Strategy Modelling

Archetype	Characteristics	Example
Aggressive	Early stop, 2-stop preferred	Red Bull 2022–23
Conservative	Late stop, 1-stop preferred	Mercedes 2021
Reactive	Responds to leader within 2 laps	Most midfield
Opportunistic	SC-triggered, flexible	Alpine, McLaren

2-player Nash equilibrium for undercut:

When gap falls below threshold, solve 2x2 simultaneous game (you pit / stay vs opponent pit / stay). Payoffs from Layer 4 warm-up model, degradation curve, pit loss samples. Closed-form Nash equilibrium mixing probability reported as undercut confidence.

Step 6 — Traffic Model

Situation	Lap Time Penalty
Clean air (gap > 2s)	0.0s
DRS range (0.5 < gap ≤ 1.0s)	-0.1s (DRS benefit)
DRS train (gap ≤ 0.5s)	+0.3–0.5s
Midfield pack (gap ≤ 0.3s)	+0.5–1.0s

Post-hoc correction: optimise assuming clean air, apply traffic penalty by predicted position, iterate once. Tyre energy dissipation multiplied by $\alpha_{\text{traffic}} = 1.08\text{--}1.12$ during traffic laps.

Step 7 — Undercut / Overcut Viability

$$W_{\text{undercut}}(j) = t_{\text{opp_degraded}}(j) - t_{\text{out_fresh}}(j) - t_{\text{pitloss}}(j) - g_{\text{current}}(j)$$

$$P(\text{undercut viable}) = (1/N_{\text{MC}}) * \sum_j 1[W_{\text{undercut}}(j) > 0]$$

$$g^* = \text{gap s.t. } P(\text{undercut viable} \mid \text{gap} = g^*) = 0.70 \text{ [decision threshold]}$$

g^* recomputed every 5 laps as tyre states evolve. Overcut viability symmetric — stays out while opponent loses more time on worn tyres than the pit stop costs.

Step 8 — Receding Horizon Dynamic Programming

$$s_n = (E_{\text{tyre}}, \text{position}, \text{gap}, \text{compound}, \text{stint}_n, \text{SC_status})$$

$$V(s_n) = \min_a E[\sum_{k=n}^N t_{\text{lap}}(s_k, a_k) + t_{\text{pitloss}} * 1[a_k = \text{pit}]]$$

$$V(s_k) = \min_{a_k} [t_{\text{lap}}(s_k, a_k) + E[V(s_{k+1}) \mid s_k, a_k]]$$

Re-solved every 5 laps. Immediate re-solve triggered by: SC deployment/end, unexpected opponent pit, cliff probability > 0.40, gap crossing g^* .

Step 9 — Aggregation & Scoring

$$\text{Score}_s = w_1 * T_{s_mean} + w_2 * \text{CVaR}_{0.8}(T_s) \text{ [} w_1=0.7, w_2=0.3 \text{]}$$

10,000 scenarios. Convergence verified by checking strategy ranking stability between 5,000 and 10,000 runs.

Layer 5 Final Output

Output	Form
Optimal strategy	Compound sequence + pit laps, ranked by Score
Expected race time	Distribution over 10,000 scenarios
Risk ranking	CVaR-adjusted strategy scores
Undercut viability	P(undercut viable) + g^* threshold
Overcut viability	P(overcut viable), updated every 5 laps
SC robustness R_s	P(strategy remains optimal SC at any lap)
Cliff warning	P(cliff within k laps) from Layer 4 survival model
Scenario branches	No-SC optimal vs SC-at-peak-impact optimal

Documented Assumptions

- Laps 1–3 modelled empirically, not from physics
- Traffic penalty calibrated from historical clean-air vs traffic lap time deltas
- Opponent archetypes fitted from 2018–2024 — may not reflect current team strategies
- Unsafe release probability: 0.005–0.02 per stop (team specific)
- SC lap multipliers: 3.0x laps 1–3, 1.0x mid-race, 1.3x final 20%
- Nash equilibrium assumes rational opponent with full information
- 10,000 scenarios sufficient for strategy ranking convergence
- Receding horizon re-solve every 5 laps or on trigger events

9. Validation Strategy

Per-Layer Validation

Layer	Validation Method
L1 — Aero	Cd/Cl posteriors vs published F1 aero configs and DRS delta_Cd literature range
L2 — Alignment	Sector time recovery error < 0.05s; transitivity check epsilon_ABC < 0.5m
L3 — Racing Line	Predicted apex speed within ±5 km/h of fastest qualifier telemetry
L4 — Tyre	Degradation curve vs multi-stint race data; cliff timing vs historical cliff events
L5 — Strategy	SC timing distribution vs historical; pit stop distribution vs FastF1 records

Counterfactual Backtesting Circuits

Validate on strategy-dominant circuits where the best strategy almost always wins regardless of SC timing:

- Monaco 2022 — minimal overtaking, pure strategy race
- Singapore 2023 — street circuit, strategy decisive
- Hungary 2022 — low overtaking, strategy dominant

SC Robustness Metric

$$R_s = P(\text{strategy } s \text{ remains optimal} \mid \text{SC occurs at any lap})$$

Strategies with $R_s > 0.65$ flagged as SC-robust recommendations.

Driver Performance Margin

Gap between predicted optimal speed profile and actual driver speed is not a validation failure — it quantifies how much performance the driver left available due to tyre management, traffic, or setup compromise. This is an additional useful output.

10. Documented Assumptions Master List

All assumptions are documented here for transparency. These are the known approximations the model makes due to data limitations. Any result should be interpreted in light of these.

Layer 1 — Aero

Assumption	Value / Notes
Car mass	798 kg with driver, ± 2 kg
Fuel burn	Linear, 1.8 kg/lap
Lap time fuel sensitivity	0.03 s/kg
Reference area A	1.5 m ² (published F1 approximation)
Engine power	Approximated from RPM/gear ($\pm 8\%$)
Rolling resistance	Prior 500–600 N ($\pm 15\%$)
DRS delta_Cd	0.08–0.12 (literature range, calibration bound)
Street circuit params	Inherited from similar config race, sigma widened x1.5

Layer 2 — Alignment

Assumption	Value / Notes
Circuit length	Official F1 figure, accurate to ± 1 m
Max timing offset	2.0s inter-driver upper bound for DTW window
Dead reckoning error	~ 0.35 m per 250ms GPS interval at max speed
Weather GP — track temp	Length scale l = 5 minutes
Weather GP — air temp	Length scale l = 15 minutes
Distance grid	1m spacing sufficient for Layer 3 curvature

Layer 3 — Racing Line

Assumption	Value / Notes
Racing line type	Geometric minimum curvature — approximation to minimum time
Vehicle model	Point mass — weight transfer is second-order correction
mu calibration	Per circuit from qualifying; may not reflect wet conditions
OSM boundaries	Accurate to ± 1 m of true track edge
Track evolution	Late-session laps assumed fully evolved

Layer 4 — Tyre

Assumption	Value / Notes
Tyre model	Energy-based wear proxy, not full thermal simulator

Stress multiplier	alpha_dominant = 1.15–1.25 for most-stressed tyre
Cliff threshold	>1.5s single-lap jump not explained by external events
Cold grip	mu_cold = 0.75 * mu_opt (literature approximation)
Compound hardness	C1=5, C2=4, C3=3, C4=2, C5=1 index
Cooling	Modelled as energy retention modifier, not explicit temperature
Compound ordering	Softer compound = faster degradation (prior constraint)

Layer 5 — Strategy

Assumption	Value / Notes
Race start	Laps 1–3 empirical, not physics-modelled
Traffic penalty	Calibrated from historical clean-air vs traffic deltas
Opponent archetypes	Fitted 2018–2024 — may not reflect current strategies
Unsafe release prob	0.005–0.02 per stop (team specific)
SC lap 1 multiplier	3.0x baseline SC rate
SC final 20% multiplier	1.3x baseline SC rate
Nash equilibrium	Assumes rational opponent with full information
MC convergence	10,000 scenarios; verified by 5k vs 10k stability check
DP re-solve frequency	Every 5 laps or on trigger events